Numerical Analysis

Chapter 6: Numerical Differentiation and Integration

Part 2, Numerical Integration

Department of Mathematics & Statistics
Missouri University of Science & Technology

Textbook: Classical and Modern Numerical Analysis: Theory, Methods and Practice
A. S. Ackleh, E. J. Allen, R. B. Kearfott and P. Seshaiyer
CRC Press, Boca Raton, FL, 2010

Outline

1. Numerical integration/quadrature
2. Other topics

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2 Other topics

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Basic idea:

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Definition (quadrature weights and nodes)

In

$$Q(f) = \sum_{j=0}^{m} \alpha_j f(x_j),$$

α_j $(j = 0, \cdots, m)$ are called quadrature weights and x_j $(j = 0, \cdots, m)$ are called quadrature nodes.

Introduction

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- $\sum_{j=0}^{m} \alpha_j = b - a.$

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That is, if $f(x)$ is a constant function, then the numerical quadrature is exact.

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- If $x \in [a, b]$, then

$$a \leq x \leq b \Rightarrow -\frac{b-a}{2} \leq x - \frac{a+b}{2} \leq \frac{b-a}{2} \Rightarrow -1 \leq \frac{x - \frac{a+b}{2}}{\frac{b-a}{2}} \leq 1.$$

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- Let $\hat{x} = \frac{x - \frac{a+b}{2}}{\frac{b-a}{2}}$. Then $\hat{x} \in [-1, 1]$ and $x = \frac{b-a}{2}\hat{x} + \frac{a+b}{2}$.

Introduction

- Hence

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$$\begin{aligned} J(f) &= \int_a^b f(x)\, dx \\ &= \int_{-1}^{1} f\left(\frac{b-a}{2}\hat{x} + \frac{a+b}{2}\right)\, d\left(\frac{b-a}{2}\hat{x} + a\right) \end{aligned}$$

Introduction

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$$
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J(f) &= \int_a^b f(x)\, dx \\
&= \int_{-1}^{1} f\left(\frac{b-a}{2}\hat{x} + \frac{a+b}{2}\right)\, d\left(\frac{b-a}{2}\hat{x} + a\right) \\
&= \int_{-1}^{1} f\left(\frac{b-a}{2}\hat{x} + \frac{a+b}{2}\right)\frac{b-a}{2}\, d\hat{x}.
\end{aligned}
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- Assume we already have the numerical quadrature for $\hat{f}(\hat{x})$ on $[-1, 1]$:

$$\hat{J}(\hat{f}) = \int_{-1}^1 \hat{f}(\hat{x})\, d\hat{x} \approx \hat{Q}(\hat{f}) = \sum_{j=0}^m \hat{\alpha}_j \hat{f}(\hat{x}_j).$$

Introduction

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$$\approx \quad \hat{Q}(\hat{f}) = \sum_{j=0}^{m} \hat{\alpha}_j \hat{f}(\hat{x}_j) = \sum_{j=0}^{m} \hat{\alpha}_j \frac{b-a}{2} f\left(\frac{b-a}{2}\hat{x}_j + \frac{a+b}{2}\right).$$

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$$\approx \quad \hat{Q}(\hat{f}) = \sum_{j=0}^{m} \hat{\alpha}_j \hat{f}(\hat{x}_j) = \sum_{j=0}^{m} \hat{\alpha}_j \frac{b-a}{2} f\left(\frac{b-a}{2}\hat{x}_j + \frac{a+b}{2}\right).$$

- Let $\alpha_j = \hat{\alpha}_j \frac{b-a}{2}$ and $x_j = \frac{b-a}{2}\hat{x}_j + \frac{a+b}{2}$. Then

$$J(f) = \int_a^b f(x)\, dx \approx Q(f) = \sum_{j=0}^{m} \alpha_j f(x_j).$$

Introduction

- Homework: Assume the numerical quadrature for $\hat{f}(\hat{x})$ on $[0, 1]$ is

$$\hat{J}(\hat{f}) = \int_0^1 \hat{f}(\hat{x})\, d\hat{x} \approx \hat{Q}(\hat{f}) = \sum_{j=0}^{m} \hat{\alpha}_j \hat{f}(\hat{x}_j).$$

  Derive the numerical quadrature of $J(f) = \int_a^b f(x)\, dx$.

Newton-Cotes formulas

Basic idea:

- The quadrature nodes x_j $(j = 0, \cdots, m)$ are taken to be equally spaced on $[a, b]$.

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- The quadrature nodes x_j $(j = 0, \cdots, m)$ are taken to be equally spaced on $[a, b]$.
- Recall the *n*-th order approximation capability of the n-th order Lagrange polynomial approximation based on $(n+1)$-th nodes.

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- Can we use the $m+1$ quadrature nodes to construct a quadrature rule which is exact for m-th order polynomial?

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- Can we use the $m+1$ quadrature nodes to construct a quadrature rule which is exact for m-th order polynomial?
- Yes!

Newton-Cotes formulas

Recall

Theorem

For any $n+1$ distinct real numbers $x_0, x_1, x_2, \cdots, x_n$ and for $n+1$ arbitrary numbers $y_0, y_1, y_2, \cdots, y_n$, let

$$L_k(x) = \prod_{i=0, i\neq k}^{n} \frac{x - x_i}{x_k - x_i}.$$

Then there exists a unique interpolating polynomial

$$p(x) = \sum_{k=0}^{n} y_k L_k(x)$$

of degree at most n such that $y_j = p(x_j)$ $(j = 0, 1, 2, \cdots, n)$.

Newton-Cotes formulas

Recall

Definition (Lagrange polynomial approximation)

$p(x) = \sum_{k=0}^{n} f(x_k) L_k(x)$ is called the Lagrange polynomial approximation/interpolation of a given function $f(x)$.

Newton-Cotes formulas

Recall

Theorem

If $x_0, x_1, x_2, \cdots, x_n$ are $n+1$ distinct points in $[a, b]$ and $f \in C^{n+1}[a, b]$, then for each $x \in [a, b]$, there exists a number $\xi = \xi(x) \in (a, b)$ such that

$$f(x) - p(x) = R(x) = \frac{f^{(n+1)}(\xi(x))W(x)}{(n+1)!}$$

where $W(x) = \prod_{i=0}^{n}(x - x_i)$ and $p(x)$ is the Lagrange polynomial approximation/interpolation.

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- The quadrature nodes x_j $(j = 0, \cdots, m)$ are taken to be equally spaced on $[a, b]$.
- Assume $p(x)$ is the Lagrange polynomial approximation of $f(x)$ based on x_j $(j = 0, \cdots, m)$. Then

$$J(f) = \int_a^b f(x)\, dx$$

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$$\begin{aligned} J(f) &= \int_a^b f(x)\, dx \\ &= \int_a^b p(x)\, dx + \int_a^b R(x)\, dx \end{aligned}$$

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Newton-Cotes formulas

- Let $\alpha_j = \int_a^b L_j(x)\ dx\ (j = 0, 1, \cdots, m)$. Then

$$J(f) \approx Q(f) = \sum_{j=0}^{m} \alpha_j f(x_j).$$

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$$J(f) \approx Q(f) = \sum_{j=0}^{m} \alpha_j f(x_j).$$

- And

$$E(f) = J(f) - Q(f) = \int_a^b R(x)\ dx = \int_a^b \frac{f^{(m+1)}(\xi(x))W(x)}{(m+1)!}\ dx.$$

where $W(x) = \prod_{j=0}^{m}(x - x_j)$.

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- If f is a polynomial of degree $\leq m$, then $f^{(m+1)} \equiv 0 \Rightarrow E(f) = J(f) - Q(f) = 0$.

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  where $W(x) = \prod_{j=0}^{m}(x - x_j)$.

- If f is a polynomial of degree $\leq m$, then $f^{(m+1)} \equiv 0 \Rightarrow E(f) = J(f) - Q(f) = 0$.

- Hence $Q(f)$ is exact at least for polynomials of degree $\leq m$. That is, $Q(f) \equiv J(f)$ when f is a polynomial of degree $\leq m$.

Newton-Cotes formulas

- Example 3: Compute the quadrature weights of the Newton-Cotes formula on [0,1] for $x_0 = 0$, $x_1 = \frac{1}{3}$, $x_2 = \frac{2}{3}$, $x_3 = 1$.

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- Example 3: Compute the quadrature weights of the Newton-Cotes formula on [0,1] for $x_0 = 0$, $x_1 = \frac{1}{3}$, $x_2 = \frac{2}{3}$, $x_3 = 1$.
- First,

$$\alpha_0 = \int_0^1 L_0(x)\, dx = \int_0^1 \prod_{i=0, i\neq 0}^{m} \frac{x - x_i}{x_0 - x_i}\, dx$$

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$$\begin{aligned} \alpha_0 &= \int_0^1 L_0(x)\, dx = \int_0^1 \prod_{i=0, i\neq 0}^{m} \frac{x - x_i}{x_0 - x_i}\, dx \\ &= \int_0^1 \frac{(x - x_1)(x - x_2)(x - x_3)}{(x_0 - x_1)(x_0 - x_2)(x_0 - x_3)}\, dx \end{aligned}$$

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&= \int_0^1 \frac{(x - x_1)(x - x_2)(x - x_3)}{(x_0 - x_1)(x_0 - x_2)(x_0 - x_3)}\, dx \\
&= \int_0^1 \frac{(x - \frac{1}{3})(x - \frac{2}{3})(x - 1)}{(0 - \frac{1}{3})(0 - \frac{2}{3})(0 - 1)}\, dx
\end{aligned}$$

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&= \int_0^1 \frac{(x - x_1)(x - x_2)(x - x_3)}{(x_0 - x_1)(x_0 - x_2)(x_0 - x_3)}\, dx \\
&= \int_0^1 \frac{(x - \frac{1}{3})(x - \frac{2}{3})(x - 1)}{(0 - \frac{1}{3})(0 - \frac{2}{3})(0 - 1)}\, dx \\
&= -\frac{9}{2} \int_0^1 \left(x^3 - 2x^2 + \frac{11}{9} - \frac{2}{9} \right)\, dx = \frac{1}{8}.
\end{aligned}$$

Newton-Cotes formulas

- Similarly, $\alpha_1 = \frac{3}{8}$, $\alpha_2 = \frac{3}{8}$, and $\alpha_3 = \frac{1}{8}$.

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- Hence

$$\int_0^1 f(x)\, dx \approx \frac{1}{8} f(0) + \frac{3}{8} f(\frac{1}{3}) + \frac{3}{8} f(\frac{2}{3}) + \frac{1}{8} f(1).$$

Newton-Cotes formulas

Remark

There is another way to derive the Newton-Cotes formulas. The basic idea is to set up $m+1$ equations for α_j $(j = 0, 1, \cdots, m)$ by requiring $J(f) = Q(f)$ for $f = 1, x, x^2, \cdots, x^m$. (Independent study problem: see section 6.3.2.1 on page 336-338 of the textbook for more details).

Newton-Cotes formulas

Definition (closed Newton-Cotes formulas)

The $m+1$ point closed Newton-Cotes formulas have points $x_j = a + jh$ $(j = 0, 1, \cdots, m)$ where $h = \frac{b-a}{m}$.

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Definition (open Newton-Cotes formulas)

The $m+1$ point open Newton-Cotes formulas have points $x_j = a + (j+1)h$ $(j = 0, 1, \cdots, m)$ where $h = \frac{b-a}{m+2}$.

Newton-Cotes formulas

- Example 4: two examples for closed and open Newton-Cotes formulas respectively.

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Newton-Cotes formulas

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&= \int_a^b \frac{x-b}{a-b}\, dx = \frac{1}{a-b}\left(\frac{1}{2}x^2 - bx\right)\Big|_{x=a}^{x=b} \\
&= \frac{1}{a-b}\left[\left(\frac{1}{2}b^2 - b^2\right) - \left(\frac{1}{2}a^2 - ab\right)\right].
\end{aligned}$$

Newton-Cotes formulas

- Then

$$\alpha_0 = \frac{1}{a-b}\left[-\frac{1}{2}b^2 - \frac{1}{2}a^2 + ab\right]$$

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- Similarly, $\alpha_1 = \frac{b-a}{2}$.
- Hence the trapezoid rule:

$$J(f) = \int_a^b f(x)\,dx \approx Q(f) = \frac{b-a}{2}f(a) + \frac{b-a}{2}f(b).$$

Newton-Cotes formulas

- Simpson's rule (Homework): $m = 2$, $x_0 = a$, $x_1 = \frac{a+b}{2}$, $x_2 = b$.

$$J(f) = \int_a^b f(x)\, dx$$

$$\approx \quad Q(f) = \frac{b-a}{6} f(a) + \frac{2(b-a)}{3} f(\frac{a+b}{2}) + \frac{b-a}{6} f(b).$$

Newton-Cotes formulas

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Newton-Cotes formulas

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- Middle point rule (Homework): $m = 0$, $x_0 = \frac{a+b}{2}$.

$$J(f) = \int_a^b f(x)\, dx \approx Q(f) = (b-a) f(\frac{a+b}{2}).$$

- Two point open rule (Homeowrk): $m = 1$, $x_0 = a + \frac{b-a}{3}$, $x_1 = a + \frac{2(b-a)}{3}$.

$$\begin{aligned} & J(f) = \int_a^b f(x)\, dx \\ \approx \quad & Q(f) = \frac{b-a}{2} f(a + \frac{b-a}{3}) + \frac{b-a}{2} f(a + \frac{2(b-a)}{3}). \end{aligned}$$

Newton-Cotes formulas

- Example 5: use the Newton-Cotes formulas in Examples 3 and 4 to approximate $J(f) = \int_1^2 \frac{cos\left(\frac{\pi}{4}x\right)}{sin^2\left(\frac{\pi}{4}x\right)}\, dx$.

Newton-Cotes formulas

- Example 5: use the Newton-Cotes formulas in Examples 3 and 4 to approximate $J(f) = \int_1^2 \frac{cos\left(\frac{\pi}{4}x\right)}{sin^2\left(\frac{\pi}{4}x\right)}\, dx$.
- Let $a = 1$, $b = 2$, $f(x) = \frac{cos\left(\frac{\pi}{4}x\right)}{sin^2\left(\frac{\pi}{4}x\right)}$. Then $J(f) = \int_a^b f(x)\, dx$.
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- Trapezoid rule:

$$\begin{aligned} Q(f) &= \frac{b-a}{2} f(a) + \frac{b-a}{2} f(b) \\ &= \frac{1}{2} \frac{cos\left(\frac{\pi}{4}\right)}{sin^2\left(\frac{\pi}{4}\right)} + \frac{1}{2} \frac{cos\left(\frac{\pi}{2}\right)}{sin^2\left(\frac{\pi}{2}\right)} \end{aligned}$$

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Newton-Cotes formulas

- Analytic solution: substitution rule with $u = sin\left(\frac{\pi}{4}x\right)$.

$$\begin{aligned}\int \frac{cos\left(\frac{\pi}{4}x\right)}{sin^2\left(\frac{\pi}{4}x\right)}\,dx &= \frac{4}{\pi}\int \frac{1}{u^2}\,du = -\frac{4}{\pi}\frac{1}{u} \\ &= -\frac{4}{\pi}\frac{1}{sin\left(\frac{\pi}{4}x\right)}.\end{aligned}$$

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- Hence

$$\begin{aligned}J(f) &= \int_1^2 \frac{cos\left(\frac{\pi}{4}x\right)}{sin^2\left(\frac{\pi}{4}x\right)}\,dx = -\frac{4}{\pi}\frac{1}{sin\left(\frac{\pi}{4}x\right)}\Big|_{x=1}^{x=2} \\ &= -\frac{4}{\pi}\frac{1}{sin\left(\frac{\pi}{2}\right)} + \frac{4}{\pi}\frac{1}{sin\left(\frac{\pi}{4}\right)} \\ &= -\frac{4}{\pi} + \frac{4}{\pi}\sqrt{2} \approx 0.527393087579050.\end{aligned}$$

Newton-Cotes formulas

- The error of trapezoid rule is

$$\begin{aligned} |Q(f) - J(f)| &\approx |0.707106781186548 - 0.527393087579050| \\ &= 0.179713693607498. \end{aligned}$$

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- What can we do if we have thousands of or even millions of integrations to compute?
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- One observation: all numerical quadratures are $Q(f) = \sum_{j=0}^{m} \alpha_j f(x_j)$ with different quadrature weights and nodes.

Newton-Cotes formulas

- Numerical results:

Newton-Cotes formulas	numerical integration	error
Middle point rule	0.448341529167965	0.079051558411084
Two point open rule	0.472034041575363	0.055359046003686
Trapezoid rule	0.707106781186548	0.179713693607498
Simpson's rule	0.534596613174159	0.007203525595110
Four point closed rule		

Newton-Cotes formulas

- For the four point closed rule on $[0, 1]$ in Example 3, we can use affine mapping to obtain the quadrature weights and nodes on $[a, b]$.
- Assume $\hat{Q}(\hat{f}) = \sum_{j=0}^{m} \hat{\alpha}_j \hat{f}(\hat{x}_j)$ is the quadrature formula on $[0, 1]$ and $Q(f) = \sum_{j=0}^{m} \alpha_j f(x_j)$ is the corresponding quadrature formula on $[a, b]$.
- Then by the homework of affine mapping, we have $\alpha_j = \hat{\alpha}_j(b - a)$ and $x_j = (b - a)\hat{x}_j + a$.

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- Now we analyze the error $E(f)$ in detail.

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- Recall that

$$\begin{aligned} E(f) &= \int_a^b \frac{f^{(m+1)}(\xi(x))W(x)}{(m+1)!}\,dx \\ &= \int_a^b \prod_{j=0}^{m}(x-x_j)\frac{f^{(m+1)}(\xi(x))}{(m+1)!}\,dx. \end{aligned}$$

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- Then

$$\begin{aligned} |E(f)| &\leq \int_a^b \left|\prod_{j=0}^{m}(x-x_j)\right| \left|\frac{f^{(m+1)}(\xi(x))}{(m+1)!}\right| dx \\ &\leq \frac{\left\|f^{(m+1)}\right\|_\infty}{(m+1)!}\int_a^b \left|\prod_{j=0}^{m}(x-x_j)\right| dx. \end{aligned}$$

Newton-Cotes formulas

- Consider the following change of variables:

$$z = \frac{x - a}{b - a}, z_i = \frac{x_i - a}{b - a}.$$

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- Hence

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- Let $H = b - a$ and $\beta^* = \int_0^1 \left|\prod_{j=0}^m (z - z_j)\right| dz$. Then

$$|E(f)| \leq H^{m+2}\beta^* \frac{\left\|f^{(m+1)}\right\|_\infty}{(m+1)!}.$$

Newton-Cotes formulas

Remark

The Newton-Cotes formula with $m+1$ nodes have accuracy order of at least $m+2$.

Newton-Cotes formulas

- Example 6: Analyze the error of trapezoid rule: $m = 1$, $x_0 = a$, $x_1 = b$.

$$J(f) = \int_a^b f(x)\, dx \approx Q(f) = \frac{b-a}{2} f(a) + \frac{b-a}{2} f(b).$$

Newton-Cotes formulas

- Example 6: Analyze the error of trapezoid rule: $m = 1$, $x_0 = a$, $x_1 = b$.

$$J(f) = \int_a^b f(x)\, dx \approx Q(f) = \frac{b-a}{2} f(a) + \frac{b-a}{2} f(b).$$

- Recall that

$$E(f) = \int_a^b \prod_{j=0}^{m} (x - x_j) \frac{f^{(m+1)}(\xi(x))}{(m+1)!}\, dx.$$

Newton-Cotes formulas

- Hence for the trapezoid rule, we have

$$|E(f)| = \left| \int_a^b (x-a)(x-b) \frac{f''(\xi(x))}{2} \, dx \right|$$

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- Hence for the trapezoid rule, we have

$$\begin{aligned} |E(f)| &= \left| \int_a^b (x-a)(x-b) \frac{f''(\xi(x))}{2} \, dx \right| \\ &\leq \frac{\|f''\|_\infty}{2} \int_a^b |(x-a)(x-b)| \, dx \end{aligned}$$

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Newton-Cotes formulas

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$$
\begin{aligned}
|E(f)| &= \left| \int_a^b (x-a)(x-b) \frac{f''(\xi(x))}{2} \, dx \right| \\
&\leq \frac{\|f''\|_\infty}{2} \int_a^b |(x-a)(x-b)| \, dx \\
&= -\frac{\|f''\|_\infty}{2} \int_a^b (x-a)(x-b) \, dx \\
&= \frac{\|f''\|_\infty}{12} (b-a)^3 \\
&= \frac{\|f''\|_\infty}{12} H^3.
\end{aligned}
$$

Newton-Cotes formulas

- Example 7: Analyze the error of middle point rule: $m = 0$, $x_0 = \frac{a+b}{2}$.

$$J(f) = \int_a^b f(x)\ dx \approx Q(f) = (b-a)f(\frac{a+b}{2}).$$

Newton-Cotes formulas

- Example 7: Analyze the error of middle point rule: $m = 0$, $x_0 = \frac{a+b}{2}$.

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- It's expected that the error is second order since $m = 0$.

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- It's expected that the error is second order since $m = 0$.
- However, by using Taylor expansion, a third order accuracy can be shown for the middle point rule.
- First, by Taylor expansion, we get

$$f(x) = f(\frac{a+b}{2}) + f'(\frac{a+b}{2})(x - \frac{a+b}{2}) + \frac{1}{2} f''(\xi)(x - \frac{a+b}{2})^2.$$

Newton-Cotes formulas

- Then

$$\int_a^b f(x)\ dx = \int_a^b f(\frac{a+b}{2})\ dx + \int_a^b f'(\frac{a+b}{2})(x - \frac{a+b}{2})\ dx + \int_a^b \frac{1}{2} f''(\xi)(x - \frac{a+b}{2})^2\ dx.$$

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- Note that

$$\int_a^b (x - \frac{a+b}{2})\ dx = \left[\frac{1}{2}x^2 - \frac{a+b}{2}x\right] |_{x=a}^{x=b}$$

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$$\int_a^b (x-\frac{a+b}{2})\ dx = \left[\frac{1}{2}x^2 - \frac{a+b}{2}x\right]|_{x=a}^{x=b}$$
$$= \left[\frac{1}{2}b^2 - \frac{a+b}{2}b\right] - \left[\frac{1}{2}a^2 - \frac{a+b}{2}a\right]$$

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- Note that

$$\begin{aligned}\int_a^b (x-\frac{a+b}{2})\ dx &= \left[\frac{1}{2}x^2 - \frac{a+b}{2}x\right]|_{x=a}^{x=b} \\ &= \left[\frac{1}{2}b^2 - \frac{a+b}{2}b\right] - \left[\frac{1}{2}a^2 - \frac{a+b}{2}a\right] \\ &= \left(-\frac{1}{2}ab\right) - \left(-\frac{1}{2}ab\right)\end{aligned}$$

Newton-Cotes formulas

- Then

$$\int_a^b f(x)\,dx = \int_a^b f(\frac{a+b}{2})\,dx + \int_a^b f'(\frac{a+b}{2})(x-\frac{a+b}{2})\,dx + \int_a^b \frac{1}{2}f''(\xi)(x-\frac{a+b}{2})^2\,dx.$$

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$$\begin{aligned}\int_a^b (x-\frac{a+b}{2})\,dx &= \left[\frac{1}{2}x^2 - \frac{a+b}{2}x\right]\Big|_{x=a}^{x=b} \\ &= \left[\frac{1}{2}b^2 - \frac{a+b}{2}b\right] - \left[\frac{1}{2}a^2 - \frac{a+b}{2}a\right] \\ &= \left(-\frac{1}{2}ab\right) - \left(-\frac{1}{2}ab\right) \\ &= 0.\end{aligned}$$

Newton-Cotes formulas

- Similarly,

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- Hence

$$\int_a^b f(x) \, dx = \int_a^b f(\frac{a+b}{2}) \, dx + \int_a^b f'(\frac{a+b}{2})(x - \frac{a+b}{2}) \, dx$$

Newton-Cotes formulas

- Similarly,

$$\int_a^b (x - \frac{a+b}{2})^2 \, dx = \frac{1}{12}(b-a)^3.$$

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$$\int_a^b f(x) \, dx = \int_a^b f(\frac{a+b}{2}) \, dx + \int_a^b f'(\frac{a+b}{2})(x - \frac{a+b}{2}) \, dx + \int_a^b \frac{1}{2} f''(\xi)(x - \frac{a+b}{2})^2 \, dx$$

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- Hence

$$\begin{aligned}\int_a^b f(x) \, dx &= \int_a^b f(\frac{a+b}{2}) \, dx + \int_a^b f'(\frac{a+b}{2})(x - \frac{a+b}{2}) \, dx \\ &\quad + \int_a^b \frac{1}{2} f''(\xi)(x - \frac{a+b}{2})^2 \, dx \\ &= (b-a) f(\frac{a+b}{2}) + \frac{1}{24}(b-a)^3 f''(\xi).\end{aligned}$$

Newton-Cotes formulas

Remark

If m is odd, the Newton-Cotes formula with $m+1$ nodes has accuracy order of $m+2$ and is exact for polynomials of degree m. If m is even, the Newton-Cotes formula with $m+1$ nodes has accuracy order of $m+3$ and is exact for polynomials of degree $m+1$. (Independent study problem: see Remark 6.4 on page 341-342 of the textbook for more details.)

Newton-Cotes formulas

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If m is odd, the Newton-Cotes formula with $m+1$ nodes has accuracy order of $m+2$ and is exact for polynomials of degree m. If m is even, the Newton-Cotes formula with $m+1$ nodes has accuracy order of $m+3$ and is exact for polynomials of degree $m+1$. (Independent study problem: see Remark 6.4 on page 341-342 of the textbook for more details.)

Remark

There is another way to derive the Newton-Cotes formula, which is to choose the quadrature weights α_j such that the quadrature formula $Q(f) = \sum_{j=0}^{m} \alpha_j f(x_j)$ is exact for polynomials of degree $\leq m$. (Independent study problem: see section 6.3.2.1 on page 336-338 for more details.)

Gaussian quadrature

Basic idea:

- Use $Q(f) = \sum_{j=0}^{m} \alpha_j f(x_j)$ to approximate $J(f) = \int_a^b f(x)\ dx$.

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- How many equations are needed to uniquely determine α_j and x_j ($j = 0, \cdots, m$)? 2(m+1).

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- How to set up the $2(m+1)$ equations?

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- How many equations are needed to uniquely determine α_j and x_j ($j = 0, \cdots, m$)? 2(m+1).
- How to set up the $2(m+1)$ equations?
- Require the quadrature formula to be exact for $f(x) = 1, x, x^2, \cdots, x^{2m+1}$.

Gaussian quadrature

- Example 8: Find the Gaussian quadrature of two nodes ($m = 1$) for $\int_{-1}^{1} f(x)\, dx$.

Gaussian quadrature

- Example 8: Find the Gaussian quadrature of two nodes $(m = 1)$ for $\int_{-1}^{1} f(x)\ dx$.
- $\int_{-1}^{1} f(x)\ dx \approx Q(f) = \sum_{j=0}^{m} \alpha_j f(x_j) = \alpha_0 f(x_0) + \alpha_1 f(x_1)$.

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- $\alpha_0 f(x_0) + \alpha_1 f(x_1) = \int_{-1}^{1} f(x)\ dx$ for $f(x) = 1, x, x^2, x^3$.

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- $\alpha_0 f(x_0) + \alpha_1 f(x_1) = \int_{-1}^{1} f(x)\ dx$ for $f(x) = 1, x, x^2, x^3$.
- For $f(x) = 1$: $\alpha_0 + \alpha_1 = \int_{-1}^{1} 1\ dx = 2$.

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- For $f(x) = 1$: $\alpha_0 + \alpha_1 = \int_{-1}^{1} 1\ dx = 2$.
- For $f(x) = x$: $\alpha_0 x_0 + \alpha_1 x_1 = \int_{-1}^{1} x\ dx = 0$.

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- $\alpha_0 f(x_0) + \alpha_1 f(x_1) = \int_{-1}^{1} f(x)\ dx$ for $f(x) = 1, x, x^2, x^3$.
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- For $f(x) = x^2$: $\alpha_0 x_0^2 + \alpha_1 x_1^2 = \int_{-1}^{1} x^2\ dx = \frac{2}{3}$.

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- For $f(x) = x^3$: $\alpha_0 x_0^3 + \alpha_1 x_1^3 = \int_{-1}^{1} x^3\ dx = 0$.

Gaussian quadrature

- Hence

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\alpha_0 + \alpha_1 &= 2, \\
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There are different techniques to solve this nonlinear system.

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- If $\alpha_1 = 0$, then there is no solution for the system.

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Gaussian quadrature

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- $\alpha_0 x_0 + \alpha_1 x_1 = 0 \Rightarrow \alpha_0 x_0 = -\alpha_1 x_1 \Rightarrow \frac{\alpha_0}{\alpha_1} = -\frac{x_1}{x_0}$.
- $\alpha_0 x_0^3 + \alpha_1 x_1^3 = 0 \Rightarrow \alpha_0 x_0^3 = -\alpha_1 x_1^3 \Rightarrow \frac{\alpha_0}{\alpha_1} = -\left(\frac{x_1}{x_0}\right)^3$.
- Then $-\frac{x_1}{x_0} = -\left(\frac{x_1}{x_0}\right)^3 \Rightarrow \frac{x_1}{x_0} = 0, 1, -1$.

Gaussian quadrature

- If $\frac{x_1}{x_0} = 0 \Rightarrow x_1 = 0$, then there is no solution for the system.

Gaussian quadrature

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$$\begin{aligned} \alpha_0 + \alpha_1 &= 2, \\ \alpha_0 x_0 - \alpha_1 x_0 &= 0, \\ \alpha_0 x_0^2 + \alpha_1 x_0^2 &= \frac{2}{3}, \end{aligned}$$

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- Hence the system becomes

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- Since $x_0 \neq 0$, then

$$\alpha_0 + \alpha_1 = 2, \ \alpha_0 - \alpha_1 = 0, \ \alpha_0 x_0^2 + \alpha_1 x_0^2 = \frac{2}{3}.$$

Gaussian quadrature

- $\alpha_0 + \alpha_1 = 2,\ \alpha_0 - \alpha_1 = 0 \Rightarrow \alpha_0 = 1,\ \alpha_1 = 1.$

Gaussian quadrature

- $\alpha_0 + \alpha_1 = 2,\ \alpha_0 - \alpha_1 = 0 \Rightarrow \alpha_0 = 1,\ \alpha_1 = 1.$
- Then $\alpha_0 x_0^2 + \alpha_1 x_0^2 = \frac{2}{3} \Rightarrow 2x_0^2 = \frac{2}{3} \Rightarrow x_0 = \pm\frac{1}{\sqrt{3}}.$

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- Then $x_1 = -x_0 = \mp\frac{1}{\sqrt{3}}.$

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- Then $x_1 = -x_0 = \mp\frac{1}{\sqrt{3}}.$
- Therefore, $\alpha_0 = 1,\ \alpha_1 = 1,\ x_0 = -\frac{1}{\sqrt{3}},\ x_1 = \frac{1}{\sqrt{3}}.$

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- Then $x_1 = -x_0 = \mp\frac{1}{\sqrt{3}}.$
- Therefore, $\alpha_0 = 1,\ \alpha_1 = 1,\ x_0 = -\frac{1}{\sqrt{3}},\ x_1 = \frac{1}{\sqrt{3}}.$
- Finally,

$$\int_{-1}^{1} f(x)\ dx \approx Q(f) = \alpha_0 f(x_0) + \alpha_1 f(x_1) = f(-\frac{1}{\sqrt{3}}) + f(\frac{1}{\sqrt{3}}).$$

Gaussian quadrature

Remark

The matlab command "solve" may be used to obtain the analytic solution of a simple system of equations.

Gaussian quadrature

Remark

The matlab command "solve" may be used to obtain the analytic solution of a simple system of equations.

- Homework: Find the Gaussian quadrature of three nodes ($m = 2$) for $\int_{-1}^{1} f(x)\, dx$ by using the technique in Example 8.

Gaussian quadrature

- Question: Is there a general way to construct the Gauss quadrature?

Gaussian quadrature

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- Step 1: choose the roots of the orthogonal polynomials to be the quadrature nodes.
- Step 2: use Lagrange polynomial approximation to obtain the quadrature weights. This is the same as Newton-Cotes formula.

Gaussian quadrature

- Question: Is there a general way to construct the Gauss quadrature?
- Yes, orthogonal polynomials!
- Step 1: choose the roots of the orthogonal polynomials to be the quadrature nodes.
- Step 2: use Lagrange polynomial approximation to obtain the quadrature weights. This is the same as Newton-Cotes formula.
- Consider $J(f) = \int_a^b f(x)\rho(x)\,dx$ where $\rho(x)$ is a real, positive, piecewise continuous weight function on (a, b).

Gaussian quadrature

Definition (Gaussian quadrature)

$Q(f) = \sum_{j=0}^{m} \alpha_j f(x_j)$ is called a Gaussian quadrature formula if

$$\int_a^b x^k \rho(x)\, dx = \sum_{j=0}^{m} \alpha_j x_j^k \quad \Big(J(x^k) = Q(x^k) \Big)$$

for $k = 0, 1, 2, \cdots, 2m+1$. That is, the quadrature formula is exact for polynomials of degree $\leq 2m+1$.

Gaussian quadrature

Definition

The inner product of $f(x)$ and $g(x)$ with respect to $\rho(x)$ is defined by

$$(f, g) = \int_a^b f(x)g(x)\rho(x)\ dx.$$

Gaussian quadrature

Definition

The inner product of $f(x)$ and $g(x)$ with respect to $\rho(x)$ is defined by

$$(f, g) = \int_a^b f(x)g(x)\rho(x)\ dx.$$

Definition (Orthogonal functions)

Functions $f(x)$ and $g(x)$ are orthogonal to each other if $(f, g) = 0$.

Gaussian quadrature

Definition (Orthogonal polynomials)

The set of polynomials $\{p_i(x)\}_{i=0}^{\infty}$ is called orthogonal on $[a, b]$ with respect to $\rho(x)$ if $p_i(x)$ $(i = 0, 1, \cdots)$ are orthogonal mutually, that is,

$$\int_a^b p_i(x)p_j(x)\rho(x)\ dx = 0\ \ ((p_i, p_j) = 0)$$

for $i \neq j$.

Gaussian quadrature

Remark

Recall the Gram-Schmidt orthogonalization process (Section 4.2) to find the set $\{p_i(x)\}_{i=0}^{\infty}$ *of orthogonal polynomials as follows:*

$$p_0(x) = 1,$$

Gaussian quadrature

Remark

Recall the Gram-Schmidt orthogonalization process (Section 4.2) to find the set $\{p_i(x)\}_{i=0}^{\infty}$ *of orthogonal polynomials as follows:*

$$p_0(x) = 1,$$

$$p_i(x) = x^i - \sum_{k=0}^{i-1} \frac{1}{(p_k, p_k)}(x^i, p_k)p_k(x), \ \ i = 1, 2, \cdots .$$

Gaussian quadrature

Recall

Theorem

For any $n+1$ distinct real numbers $x_0, x_1, x_2, \cdots, x_n$ and for $n+1$ arbitrary numbers $y_0, y_1, y_2, \cdots, y_n$, let

$$L_k(x) = \prod_{i=0, i\neq k}^{n} \frac{x - x_i}{x_k - x_i}.$$

Then there exists a unique interpolating polynomial

$$p(x) = \sum_{k=0}^{n} y_k L_k(x)$$

of degree at most n such that $y_j = p(x_j)$ $(j = 0, 1, 2, \cdots, n)$.

Gaussian quadrature

Recall

Definition (Lagrange polynomial approximation)

$p(x) = \sum_{k=0}^{n} f(x_k) L_k(x)$ is called the Lagrange polynomial approximation/interpolation of a given function $f(x)$.

Gaussian quadrature

Recall

Theorem

If $x_0, x_1, x_2, \cdots, x_n$ are $n+1$ distinct points in $[a, b]$ and $f \in C^{n+1}[a, b]$, then for each $x \in [a, b]$, there exists a number $\xi = \xi(x) \in (a, b)$ such that

$$f(x) - p(x) = R(x) = \frac{f^{(n+1)}(\xi(x))W(x)}{(n+1)!}$$

where $W(x) = \prod_{i=0}^{n}(x - x_i)$ and $p(x)$ is the Lagrange polynomial approximation/interpolation.

Gaussian quadrature

Construction of Gaussian quadrature:

- The quadrature nodes x_j $(j = 0, \cdots, m)$ are taken to be the $m+1$ distinct roots of p_{m+1} in (a, b).

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$$J(f) = \int_a^b f(x)\rho(x)\, dx$$

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Gaussian quadrature

- Let $\alpha_j = \int_a^b L_j(x)\rho(x)\ dx\ (j = 0, 1, \cdots, m)$. Then

$$J(f) \approx Q(f) = \sum_{j=0}^{m} \alpha_j f(x_j).$$

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- And

$$E(f) = J(f) - Q(f) = \int_a^b R(x)\rho(x)\ dx = \int_a^b \frac{f^{(m+1)}(\xi(x))W(x)}{(m+1)!}\rho(x)\ dx.$$

where $W(x) = \prod_{j=0}^{m}(x - x_j)$.

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- If f is a polynomial of degree $\leq m$, then $f^{(m+1)} \equiv 0 \Rightarrow E(f) = J(f) - Q(f) = 0$.

Gaussian quadrature

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- If f is a polynomial of degree $\leq m$, then $f^{(m+1)} \equiv 0 \Rightarrow E(f) = J(f) - Q(f) = 0$.

- Hence $Q(f)$ is exact at least for polynomials of degree $\leq m$.

Gaussian quadrature

Theorem (I)

Consider the quadrature formula $Q(f) = \sum_{j=0}^{m} \alpha_j f(x_j)$ *for* $J(f) = \int_a^b f(x)\rho(x)\,dx$. $Q(f)$ *is exact for polynomials of degree* $\leq 2m+1$ *if and only if the quadrature nodes* x_j $(j = 0, 1, \cdots, m)$ *are the roots of* $p_{m+1}(x)$ *and the quadrature weights* $\alpha_j = \int_a^b L_j(x)\rho(x)\,dx$ $(j = 0, 1, \cdots, m)$.

Gaussian quadrature

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Consider the quadrature formula $Q(f) = \sum_{j=0}^{m} \alpha_j f(x_j)$ *for* $J(f) = \int_a^b f(x)\rho(x)\,dx$. $Q(f)$ *is exact for polynomials of degree* $\leq 2m+1$ *if and only if the quadrature nodes* x_j $(j = 0, 1, \cdots, m)$ *are the roots of* $p_{m+1}(x)$ *and the quadrature weights* $\alpha_j = \int_a^b L_j(x)\rho(x)\,dx$ $(j = 0, 1, \cdots, m)$.

Proof.

See pages ?-? of the textbook. (Independent study problem or take Math 6601 (Numerical Analysis)) □

Gaussian quadrature

Remark

Except the Gram-Schmidt process, there is another algebraic procedure to obtain $p_{m+1}(x)$. Then the quadrature nodes and weights can be obtained by using Theorem (I). (Independent study problem: see Remark 6.11 on page 347-348 of the textbook for more details.)

Gaussian quadrature

Theorem (II)

Let $\rho(x) > 0$ be a weight function defined on $[a, b]$ and let $\{p_k(x)\}_{k=0}^{\infty}$ be the associated sequence of orthogonal polynomials generated by the Gram-Schmidt orthogonalization process. Let $\{x_i\}_{i=0}^{m}$ and $\{\alpha_i\}_{i=0}^{m}$ be the points and weights for Gauss quadratures respectively. If $f \in C^{2m+2}[a, b]$, we have

$$\int_a^b \rho(x) f(x)\, dx - (b-a) \sum_{j=0}^{m} \alpha_j f(x_j) = \frac{f^{(2m+2)}(\xi)}{(2m+2)!} \int_a^b \rho(x) p_{m+1}^2(x)\, dx$$

where $\xi \in (a, b)$.

Gaussian quadrature

Proof.

See page 349-350 of the textbook. (Independent study problem) □

Gaussian quadrature

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Remark

The regularity requirement $f \in C^{2m+2}[a, b]$ is critical in practice!

Gaussian quadrature

Theorem (III)

The weights α_j $(j = 0, 1, \cdots, m)$ in Gaussian quadrature are all positive for any m.

Gaussian quadrature

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The weights α_j $(j = 0, 1, \cdots, m)$ in Gaussian quadrature are all positive for any m.

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See page 352 of the textbook. (Independent study problem) □

Gaussian quadrature

Recall affine mapping:

- If $x \in [a, b]$, then

$$a \le x \le b \Rightarrow -\frac{b-a}{2} \le x - \frac{a+b}{2} \le \frac{b-a}{2} \Rightarrow -1 \le \frac{x - \frac{a+b}{2}}{\frac{b-a}{2}} \le 1.$$

Gaussian quadrature

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- Let $\hat{x} = \frac{x - \frac{a+b}{2}}{\frac{b-a}{2}}$. Then $\hat{x} \in [-1, 1]$ and $x = \frac{b-a}{2}\hat{x} + \frac{a+b}{2}$.

Gaussian quadrature

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- Then

$$\begin{aligned} J(f) &= \int_a^b f(x)\, dx = \int_{-1}^1 f\left(\frac{b-a}{2}\hat{x} + \frac{a+b}{2}\right)\ d\left(\frac{b-a}{2}\hat{x} + a\right) \\ &= \int_{-1}^1 f\left(\frac{b-a}{2}\hat{x} + \frac{a+b}{2}\right)\frac{b-a}{2}\, d\hat{x}. \end{aligned}$$

Gaussian quadrature

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- Thus we only need to consider the Gauss quadrature on $[-1, 1]$.

Gaussian quadrature

- Example 9: Gauss-Legendre quadrature rules on $[-1, 1]$, which use $\rho(x) = 1$.

Gaussian quadrature

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- m=0: $x_0 = 0$; $\alpha_0 = 2$. (middle point rule)

Gaussian quadrature

- Example 9: Gauss-Legendre quadrature rules on $[-1, 1]$, which use $\rho(x) = 1$.
- m=0: $x_0 = 0$; $\alpha_0 = 2$. (middle point rule)
- m=1: $x_0 = -\frac{1}{\sqrt{3}}$, $x_1 = -x_0 = \frac{1}{\sqrt{3}}$; $\alpha_0 = \alpha_1 = 1$. (Example 8)

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- m=2: $x_0 = -\sqrt{\frac{3}{5}}$, $x_1 = 0$, $x_2 = -x_0 = \sqrt{\frac{3}{5}}$; $\alpha_0 = \alpha_2 = \frac{5}{9}$, $\alpha_1 = \frac{8}{9}$.

Gaussian quadrature

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- m=3: $x_0 = -\sqrt{\frac{3+\sqrt{4.8}}{7}}$, $x_1 = -\sqrt{\frac{3-\sqrt{4.8}}{7}}$, $x_2 = -x_1 = \sqrt{\frac{3-\sqrt{4.8}}{7}}$, $x_3 = -x_0 = \sqrt{\frac{3+\sqrt{4.8}}{7}}$; $\alpha_0 = \alpha_3 = \frac{1}{2} - \frac{1}{3\sqrt{4.8}}$, $\alpha_1 = \alpha_2 = \frac{1}{2} + \frac{1}{3\sqrt{4.8}}$.

Gaussian quadrature

- Let's derive the Gauss-Legendre quadrature rule on $[-1, 1]$ with $m = 1$.

Gaussian quadrature

- Let's derive the Gauss-Legendre quadrature rule on $[-1, 1]$ with $m = 1$.
- Recall

Theorem (I)

Consider the quadrature formula $Q(f) = \sum_{j=0}^{m} \alpha_j f(x_j)$ *for* $J(f) = \int_a^b f(x)\rho(x)\,dx$. $Q(f)$ *is exact for polynomials of degree* $\leq 2m+1$ *if and only if the quadrature nodes* x_j $(j = 0, 1, \cdots, m)$ *are the roots of* $p_{m+1}(x)$ *and the quadrature weights* $\alpha_j = \int_a^b L_j(x)\rho(x)\,dx$ $(j = 0, 1, \cdots, m)$.

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- Since $m = 1$, we first need to obtain the orthogonal polynomials $p_0(x), p_1(x), p_2(x)$ with respect to the weight function $\rho(x) = 1$.

Gaussian quadrature

- Gram-Schmidt process for $u_1 = 1, u_2 = x, u_3 = x^2$ with respect to the weight function $\rho(x) = 1$:

$$\begin{aligned} v_1 &= u_1 = 1, \\ \|v_1\|^2 &= (v_1, v_1) = \int_{-1}^{1} 1 \, dx = 2, \end{aligned}$$

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\|v_2\|^2 &= (v_2, v_2) = \int_{-1}^{1} x^2 \, dx = 2/3, \\
v_3 &= u_3 - \frac{(u_3, v_1)}{(v_1, v_1)} v_1 - \frac{(u_3, v_2)}{(v_2, v_2)} v_2
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v_3 &= u_3 - \frac{(u_3, v_1)}{(v_1, v_1)} v_1 - \frac{(u_3, v_2)}{(v_2, v_2)} v_2 \\
&= x^2 - \frac{1}{2} \int_{-1}^{1} x^2 \, dx - \frac{x}{2/3} \int_{-1}^{1} x^3 \, dx
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Gaussian quadrature

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v_3 &= u_3 - \frac{(u_3, v_1)}{(v_1, v_1)} v_1 - \frac{(u_3, v_2)}{(v_2, v_2)} v_2 \\
&= x^2 - \frac{1}{2} \int_{-1}^{1} x^2 \, dx - \frac{x}{2/3} \int_{-1}^{1} x^3 \, dx \\
&= x^2 - \frac{1}{2} \cdot \frac{2}{3} - \frac{x}{2/3} \cdot 0 = x^2 - \frac{1}{3}.
\end{aligned}
$$

Gaussian quadrature

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Gaussian quadrature

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- So $x_0 = -\frac{\sqrt{3}}{3}$, $x_1 = \frac{\sqrt{3}}{3}$.

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$$\begin{aligned} \alpha_0 &= \int_{-1}^{1} L_0(x)\rho(x)\, dx = \int_{-1}^{1} \frac{x - x_1}{x_0 - x_1}\, dx \\ &= \int_{-1}^{1} \frac{x - \frac{\sqrt{3}}{3}}{-\frac{\sqrt{3}}{3} - \frac{\sqrt{3}}{3}}\, dx = \left(-\frac{\sqrt{3}}{4}x^2 + \frac{1}{2}x \right) \Big|_{x=-1}^{x=1} \end{aligned}$$

Gaussian quadrature

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\end{aligned}$$

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&= 1.
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- Similarly, $\alpha_1 = 1$.

Gaussian quadrature

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\end{aligned}$$

- Similarly, $\alpha_1 = 1$.
- Hence, for $m = 1$: $x_0 = -\frac{1}{\sqrt{3}}$, $x_1 = -x_0 = \frac{1}{\sqrt{3}}$; $\alpha_0 = \alpha_1 = 1$.

Gaussian quadrature

- Example 10: use the Gauss formulas in Example 9 to approximate $J(f) = \int_1^2 \frac{cos\left(\frac{\pi}{4}x\right)}{sin^2\left(\frac{\pi}{4}x\right)}\, dx$, which is the same integral as in Example 5.

Gaussian quadrature

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- Let $a = 1$, $b = 2$, $f(x) = \frac{cos\left(\frac{\pi}{4}x\right)}{sin^2\left(\frac{\pi}{4}x\right)}$. Then $J(f) = \int_a^b f(x)\, dx$.

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- Let $a = 1$, $b = 2$, $f(x) = \frac{cos\left(\frac{\pi}{4}x\right)}{sin^2\left(\frac{\pi}{4}x\right)}$. Then $J(f) = \int_a^b f(x)\ dx$.
- Recall the analytic solution from Example 5:

$$J(f) = -\frac{4}{\pi} + \frac{4}{\pi}\sqrt{2} \approx 0.527393087579050.$$

Practice in Matlab for Gaussian quadrature

- Recall the numerical results of Example 5:

Newton-Cotes formulas	numerical integration	error
Middle point rule	0.448341529167965	0.079051558411084
Two point open rule	0.472034041575363	0.055359046003686
Trapezoid rule	0.707106781186548	0.179713693607498
Simpson's rule	0.534596613174159	0.007203525595110
Four point closed rule	0.530802226478159	0.003409138899110

Practice in Matlab for Gaussian quadrature

- For the Gaussian quadratures on $[-1, 1]$ in Example 9, we can use affine mapping to obtain the quadrature weights and nodes on $[a, b]$.

Practice in Matlab for Gaussian quadrature

- For the Gaussian quadratures on $[-1, 1]$ in Example 9, we can use affine mapping to obtain the quadrature weights and nodes on $[a, b]$.
- Assume $\hat{Q}(\hat{f}) = \sum_{j=0}^{m} \hat{\alpha}_j \hat{f}(\hat{x}_j)$ is the quadrature formula on $[-1, 1]$ and $Q(f) = \sum_{j=0}^{m} \alpha_j f(x_j)$ is the corresponding quadrature formula on $[a, b]$.

Practice in Matlab for Gaussian quadrature

- For the Gaussian quadratures on $[-1, 1]$ in Example 9, we can use affine mapping to obtain the quadrature weights and nodes on $[a, b]$.
- Assume $\hat{Q}(\hat{f}) = \sum_{j=0}^{m} \hat{\alpha}_j \hat{f}(\hat{x}_j)$ is the quadrature formula on $[-1, 1]$ and $Q(f) = \sum_{j=0}^{m} \alpha_j f(x_j)$ is the corresponding quadrature formula on $[a, b]$.
- Then by the lecture of affine mapping, we have $\alpha_j = \hat{\alpha}_j \frac{b-a}{2}$ and $x_j = \frac{b-a}{2} \hat{x}_j + \frac{a+b}{2}$.

Practice in Matlab for Gaussian quadrature

- Numerical results:

Newton-Cotes formulas	numerical integration	error
Middle point rule	0.448341529167965	0.079051558411084
Two point open rule	0.472034041575363	0.055359046003686
Trapezoid rule	0.707106781186548	0.179713693607498
Simpson's rule	0.534596613174159	0.007203525595110
Four point closed rule	0.530802226478159	0.003409138899110
One point Gaussian	0.448341529167965	0.079051558411084
Two point Gaussian	0.522795202310018	0.004597885269032
Three point Gaussian	0.527191770561494	0.000201317017555
Four point Gaussian	0.527385259852249	0.000007827726801

Outline

1 Numerical integration/quadrature

2 Other topics

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- Monte Carlo methods (Section 6.3.6): sampling.

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